

AI-Driven Intelligent Freight Quotation & Risk Platform

A production-ready quotation and risk engine combining multi-modal routing, machine learning rate regression, marine weather telemetry, customs compliance RAG, and geospatial catchment validation.

Project: Infosys Industry Project (M1–M3+)	Author / Engineer: Naresh Ramavath	Status: Production Ready (100% Tested)
Frontend: React 18, Vite 8, Leaflet, Chart.js	Backend: Django REST Framework, Python 3.11+	Database: MongoDB & Document Models

1. System Overview & Problem Formulation

Commercial freight forwarding has long suffered from slow manual rate discovery, fragmented rate sheets, and high pricing uncertainty. Standard international quote turnaround times typically span 24 to 48 hours. Furthermore, static rate cards fail to account for dynamic port congestion, changing bunker fuel prices, localized marine weather hazards, and strict customs documentation requirements.

This platform replaces manual workflows with an automated multi-agent architecture. The engine ingests origin/destination nodes, cargo parameters, and Incoterms, and produces an itemized, defensible quotation in under 7 seconds. The pricing model enforces lane margin floors, cross-references live weather along oceanic waypoints, conducts automated customs tariff compliance checks, and verifies door-to-port drayage boundaries.

2. Autonomous 6-Agent Service Architecture

The core quotation pipeline orchestrates six specialized micro-services that execute concurrently to evaluate the consignment:

Service / Agent	Core Responsibility	Inputs / Telemetry Sources	Outputs Generated
Orchestrator	Coordinates concurrent agent execution, merges sub-task outputs, enforces status transitions, and finalizes quotation objects.	Client enquiry payload, auth token	Consolidated quote record, booking ID
Route Agent	Calculates geodesic distances, waypoint trajectories, mode-specific transit durations, and verifies UN/LOCODE port pairs.	Port coordinates, transit tables, OpenStreetMap	Route GeoJSON, transit days, nautical miles
Pricing Agent	Executes hybrid pricing: deterministic rate cards (M1) and Gradient Boosted ML regression (M2) with lane margin protection.	Rate cards, container dimensions, weight, fuel indices	Itemized buy rate, margin %, final sell rate
Weather Agent	Queries live meteorological conditions along shipping lanes. Evaluates wave height, wind speed, oceanic swell, and storm paths.	Open-Meteo Marine API, GFS/Copernicus data	Weather risk score (0–100), waypoint reports
Customs Agent	Performs semantic vector search over HS Code tariffs, validates country import rules, and audits mandatory shipping documents.	Harmonized Tariff System (HTS), FAISS vectors	Customs score, missing docs list, hold flags
Risk Agent	Executes a 5-factor Multi-Criteria Decision Analysis (MCDA) model to assign an aggregate risk score and policy decision.	Weather, customs, congestion, carrier, geopolitical	Overall risk index, policy action (Auto/Review/Hold)

3. Milestone Deliverables & Technical Formulations

Milestone 1: Core Foundation & Deterministic Pricing Engine

- Multi-Modal Coverage:** Ocean FCL (20' Standard, 40' Standard, 40' High-Cube, 45' High-Cube), Ocean LCL, Air Cargo, Express, and Inland Rail/Road Depots.
- Incoterms 2020 Compliance:** Cost and risk demarcations strictly implemented across EXW, FCA, FOB, CFR, CIF, DAP, and DDP.
- Itemized Cost Breakdown:** Clear division between base ocean freight, fuel surcharges (BAF), origin terminal handling (THC), destination charges, and customs fees.

Cost Component	Calculation Logic / Formula	Application Rule
Base Freight Rate	Distance (NM/KM) × Base Rate/Unit × Weight Factor	Mandatory linehaul charge
Bunker Adjustment (BAF)	Base Freight × Fuel Surcharge Rate (7.5% – 15.0%)	Indexed against marine fuel prices
Terminal Handling (THC)	Port Authority Scheduled Tariff per Container / TEU	Origin & Destination port terminals
Documentation Fee	Fixed Administrative Charge per Bill of Lading (B/L)	Export filing & manifest issuance
Marine Insurance	Declared Cargo Commercial Value × 0.35% (ICC-A Terms)	Applied on CIF/CIP or shipper opt-in

Milestone 2: Machine Learning Rate Regression & Margin Intelligence

- Gradient Boosted ML Model:** Trained on historic shipping transactions to identify non-linear lane demand, seasonal capacity crunches, and carrier volume discounts.
- Commercial Margin Management:** Automated profit margin recommendations with lane-specific floor enforcement (e.g. minimum 12.0% floor on Asia–Europe lanes). Below-floor quotes automatically route to the Broker Review Queue.
- Two-Phase Rate Card Importer:** Allows freight forwarders to import, validate, and commit Excel/CSV carrier rate cards directly into MongoDB.

Milestone 3: Marine Weather Telemetry, Customs RAG & 5-Factor Risk Engine

- Live Marine Weather Integration:** Open-Meteo GFS marine data sampled along oceanic waypoints (Arabian Sea, Malacca Strait, Mediterranean, North Sea). Tracks wave heights > 4.0m, wind speeds > 35 knots, and storm tracks.
- Customs Document Compliance (RAG):** Vector similarity search over HTS code rules, identifying required trade paperwork (Bill of Lading, Commercial Invoice, Certificate of Origin, Safety Data Sheets).
- 5-Factor Multi-Criteria Decision Analysis (MCDA):** Computes an aggregate risk index based on weighted parameters:

Risk Parameter	Weight	Data Sources & Threshold Conditions Evaluated
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Marine Weather	30%	Wave height > 4.0m, wind speed > 35 kts, swell > 3.0m along trajectory waypoints
Customs Compliance	25%	Missing mandatory shipping documents, high-risk HS code tariff restrictions
Route Congestion	20%	Terminal berth wait times, bottleneck corridors (Suez Canal, Malacca, Panama Canal)
Carrier Reliability	15%	Carrier historical on-time performance (OTP) and container roll rates
Geopolitical Security	10%	Maritime security advisories, regional sanctions, and piracy transit corridors

4. Operational Portals & Status Lifecycle State Machines

The system delivers dedicated interfaces tailored to each logistics stakeholder, integrated with a strict status transition model:

Stakeholder Portal	Target User Group	Core Functional Capabilities & Workflows
Customer Portal	Shippers, Importers, Exporters	Self-service enquiry submission, instant quote generation, door-to-port route mapping, address book management, 1-click booking accept/reject, PDF export.
Freight Agent Desk	Brokers & Forwarders	Quotation review queue, margin override controls, spot rate bargaining desk, carrier rate sheet ingestion, shipper volume analytics.
Customs Officer Portal	Border Authorities & Inspectors	Shipment document inspection desk, HS code tariff audit, regulatory hold flags, digital customs sign-off modal with audit trail.
Admin Command Center	Operations Directors & Admins	Real-time 6-agent health monitor, master data CRUD (ports, carriers, routes, rates), user RBAC, system event logs.

Status Lifecycle State Machine Specifications (server/quotes/lifecycle.py):

- **Shipment Lifecycle (6 States):** DRAFT → SUBMITTED → PROCESSING → ANALYZED → QUOTED → CLOSED / CANCELLED
- **Quotation Lifecycle (7 States):** DRAFT → GENERATED → PENDING_REVIEW → APPROVED → SENT → ACCEPTED / REJECTED / EXPIRED
- **State Transition Invariant:** The backend enforces strict finite-state validation. Illegal jumps (e.g. attempting to mark an unapproved quote as ACCEPTED) return HTTP 400 Bad Request.

5. Geospatial Proximity & Catchment Engine

To prevent severe operational and pricing discrepancies, the platform implements a specialized **Geospatial Catchment Engine** (client/src/utils/geoProximity.js):

- **The Catchment Problem:** Shippers frequently choose a departure port (e.g. *Nhava Sheva / Mumbai*) but select a pickup facility located across the continent (e.g. *Chennai, 1,030 km away*), leading to inaccurate drayage estimates and scheduling failures.
- **Haversine & Road Distance Engine:** Calculates true geodesic distances between address coordinates and port terminals with a 15% overland detour factor.
- **Mode-Specific Drayage Limits:** Seaports (250 km max drayage), Airports (150 km bonded feeder radius), Inland Container Depots (300 km radius).
- **Proximity Mismatch Disclaimer Modal:** When an address exceeds threshold limits, the system triggers an interactive modal presenting three options: (1) **1-Click Port Switch** to the nearest operational terminal, (2) **Pick Nearby Address** within the regional port zone, or (3) **Acknowledge Long-Haul Drayage** (+₹35,000 linehaul surcharge).
- **Leaflet Door-to-Port Map Connectors:** Renders first-mile pickup connectors (green/red), linehaul ocean/air arcs, and last-mile delivery connectors (blue/red).

6. REST API Architecture & Data Persistence

The backend implements standard Django REST Framework viewsets with MongoDB persistence for quotations, rate cards, master data, and address books:

API Endpoint / URI	Method	Payload / Functional Behavior	State Machine Impact
/quotes/estimate/	POST	Submits enquiry payload (origin, destination, cargo items, Incoterm)	Initializes DRAFT quotation
/quotes/generate/	POST	Triggers 6-agent execution pipeline + 5-factor risk scoring	Transitions to GENERATED
/quotes/<id>/decision/	POST	Processes client booking decision (accept or reject)	Transitions to ACCEPTED/REJECTED
/admin/quotes/<id>/status/	PATCH	Admin/Broker manual status override with audit justification	Validates transition graph
/customs/shipments/	GET/POST	Customs inspector review queue and digital compliance sign-off	Sets customs hold / cleared
/weather/assess/	POST	Fetches live oceanic wave/wind telemetry along route waypoints	Updates weather risk score

7. Interface Standards & Build Quality Verification

- **Icon Modernization:** All informal Unicode emojis were removed and replaced with standard SVG icon components from lucide-react across all 12 platform views.
- **Client Production Build:** Vite 8.1.5 client bundle compiled cleanly with **0 syntax errors and 0 build warnings** (build time: ~9.2s).
- **Backend API Integrity:** All endpoint handlers, serialization classes, and state transitions verified against test suites.
- **Geospatial Accuracy:** Proximity engine tested against domestic and international seaport and airport coordinates.

ENGINEERING SIGN-OFF & SUBMISSION	PROJECT SPECIFICATIONS
Lead Engineer: Naresh Ramavath Role: Full-Stack & AI Systems Developer Project Title: Freight Quote Generation System Program: Infosys Industry Project Batch 2026	Milestone Scope: M1 (Core), M2 (ML), M3 (RAG/Weather), M3+ (Portals/Geo) Delivery Status: Completed & Verified Documentation Version: 3.4 Date: September 2026